

SOFTWARE REQUIREMENTS DOCUMENT

(SRD)

[Project Name — **SHISHU SNEHH HEALTHCARE APP**]

MindMatrix Industry Readiness Programme · Android Internship

Document Control

Field	Details
Document Title	Software Requirements Document (SRD) — Shishu-Sneh
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Prepared By	Mehek Munir Ahmed Tanga · Student Developer
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1. Introduction and Purpose

1.1 Purpose of This Document

This Software Requirements Document (SRD) defines the complete, precise, and testable requirements for the **Shishu-Sneh – Baby’s First Year Guide** Android application. It expands the PRD into:

- Atomic Functional Requirements (FR)
- Measurable Non-Functional Requirements (NFR)
- Detailed Use Cases
- Requirements Traceability Matrix (RTM)
- Database Schema and UI Requirements

This document acts as the primary reference for development, testing, evaluation, and stakeholder approval.

1.2 Scope

In Scope

The following functionalities are included in the project scope:

- F-01: Baby registration with name, DOB, and gender
- F-02: Growth tracking using weight and height records
- F-03: Growth chart visualization
- F-04: Automatic vaccination schedule generation
- F-05: Vaccination reminders using notifications
- F-06: Feeding and nutrition guidance
- F-07: Development milestone tracking
- F-08: Offline local storage using Room Database
- F-09: Dashboard displaying all baby health information

Out of Scope

The following functionalities are excluded:

- OOS-01: Real-time doctor consultation
- OOS-02: Hospital or government system integration
- OOS-03: Cloud storage and multi-device sync
- OOS-04: AI-based diagnosis
- OOS-05: Online payment systems
- OOS-06: Wearable device integration

1.3 Intended Audience

Reader	Purpose
Development Team	Understand exact requirements before coding
Mentor / Project Sponsor	Review and approve requirements
Tester	Create test cases from FRs and UCs
MindMatrix Evaluator	Verify implementation and evaluation criteria

2. Stakeholders and User Personas

2.1 Stakeholder Map

Stakeholder	Role	Influence
Mothers / Caregivers	Primary users	High
Community Health Workers	Assist users	Medium
Mentor	Project approval	High
Developer Team	Implementation	High

2.2 User Persona

Persona — Rural Mother

Attribute	Details
Name	Asha Devi
Age	27
Background	Rural mother with basic smartphone usage
Goal	Track baby health and vaccination schedule

Attribute	Details
Pain Points	Forgetting vaccine dates and feeding schedules
Expectation	Easy-to-use offline healthcare app

Derived Requirements

- FR-SSH-01 Baby Registration
- FR-SSH-05 Vaccination Reminders
- FR-SSH-07 Growth Charts
- NFR-PERF-01 Fast Dashboard Loading

3. System Context and Architecture Overview

Layer	Components	Role
UI Layer	XML Layouts, Activities, Fragments	User interaction
ViewModel Layer	BabyViewModel, VaccineViewModel	Business logic
Repository Layer	BabyRepository, GrowthRepository	Data mediation
Data Layer	Room Database + DAO	Local persistence
Notification Layer	AlarmManager / NotificationManager	Vaccination reminders

3.2 Architecture Constraint

No Room DAO method shall be called directly from UI classes. All database access must flow through:

UI → ViewModel → Repository → DAO

4. Functional Requirements (FR)

FR Group 1 — Baby Registration

FR ID	Requirement	Priority
FR-SSH-01	The system shall allow users to register baby details including name, date of birth, and gender.	M
FR-SSH-02	The system shall store baby details in the Baby table using Room Database.	M
FR-SSH-03	The system shall allow users to edit baby details.	M
FR-SSH-04	The system shall allow users to delete baby profiles after confirmation.	M

FR Group 2 — Growth Tracking

FR ID	Requirement	Priority
FR-SSH-05	The system shall allow users to add baby weight records.	M
FR-SSH-06	The system shall allow users to add baby height records.	M
FR-SSH-07	The system shall display graphical growth charts based on stored records.	M
FR-SSH-08	The system shall update growth charts within 2 seconds after adding new data.	M

FR Group 3 — Vaccination Management

FR ID	Requirement	Priority
FR-SSH-09	The system shall automatically generate vaccination schedules based on baby DOB.	M
FR-SSH-10	The system shall display vaccine due dates in chronological order.	M
FR-SSH-11	The system shall send reminder notifications before vaccine due dates.	M
FR-SSH-12	The system shall allow users to mark vaccines as completed.	M

FR Group 4 — Feeding and Milestone Tracking

FR ID	Requirement	Priority
FR-SSH-13	The system shall provide feeding guidance based on baby age.	M
FR-SSH-14	The system shall display milestone checklists.	M
FR-SSH-15	The system shall allow users to mark milestones as completed.	M

FR Group 5 — Dashboard and Navigation

FR ID	Requirement	Priority
FR-SSH-16	The system shall display a dashboard containing growth summary, vaccine reminders, and milestones.	M
FR-SSH-17	The system shall provide navigation between all major screens.	M
FR-SSH-18	The system shall support offline access to all saved records.	M

5. Non-Functional Requirements (NFR)

NFR ID	Category	Requirement
NFR-PERF-01	Performance	Application shall load within 2 seconds
NFR-PERF-02	Performance	Dashboard updates shall occur within 1 second
NFR-USAB-01	Usability	Users shall complete baby registration within 45 seconds
NFR-RELY-01	Reliability	Data shall persist after force close
NFR-PORT-01	Portability	Application shall support Android 8.0 and above
NFR-SEC-01	Security	User data shall remain locally stored securely
NFR-OFF-01	Offline Capability	Core app functionality shall work without internet

6. Use Case Register

UC ID	Use Case Name	Actor
UC-SSH-01	Register Baby	Mother
UC-SSH-02	Update Growth Data	Mother
UC-SSH-03	View Growth Charts	Mother
UC-SSH-04	View Vaccination Schedule	Mother

UC ID	Use Case Name	Actor
UC-SSH-05	Receive Vaccination Reminder	Mother
UC-SSH-06	View Feeding Guidance	Mother
UC-SSH-07	Track Milestones	Mother
UC-SSH-08	View Dashboard	Mother

6.2 Use Case Specification — UC-SSH-02 Update Growth Data

Field	Details
Actor	Mother / Caregiver
Trigger	User taps “Add Growth Record”
Preconditions	Baby profile exists
Main Flow	User enters weight and height → taps Save → system stores data and updates charts
Alternative Flow	User edits an existing growth record
Exception Flow	Invalid weight/height → validation error
Postconditions	Growth data saved successfully
Business Rules	Weight and height must be positive values
Acceptance Criteria	Charts update correctly within 2 seconds

7. Requirements Modelling

7.1 Use Case Diagram

INSERT: Use Case Diagram showing:

- Mother actor
- UC-SSH-01 to UC-SSH-08
- System boundary

7.2 Entity Relationship Diagram (ERD)

INSERT: ERD containing:

- Baby Table
- GrowthRecord Table
- Vaccination Table
- Milestone Table

7.3 Data Flow Diagram

INSERT: DFD showing:

User → App Screens → Room Database → Notifications / Charts

8. Data Requirements — Database Schema

Baby Table

Field	Type	Constraint
babyId	INTEGER	Primary Key
name	TEXT	NOT NULL

Field	Type	Constraint
dateOfBirth	TEXT	NOT NULL
gender	TEXT	NOT NULL

GrowthRecord Table

Field	Type	Constraint
recordId	INTEGER	Primary Key
babyId	INTEGER	Foreign Key
weight	REAL	NOT NULL
height	REAL	NOT NULL
date	TEXT	NOT NULL

Vaccination Table

Field	Type	Constraint
vaccineId	INTEGER	Primary Key
babyId	INTEGER	Foreign Key
vaccineName	TEXT	NOT NULL
dueDate	TEXT	NOT NULL
status	TEXT	NOT NULL

Milestone Table

Field	Type	Constraint
milestoneId	INTEGER	Primary Key
babyId	INTEGER	Foreign Key
weekNumber	INTEGER	NOT NULL
description	TEXT	NOT NULL
completed	BOOLEAN	Default FALSE

9. UI / UX Requirements
Colour Palette

Role	Hex Code
Primary Blue	#64B5F6
Soft Pink	#F8BBD0
Green Accent	#81C784
Background	#F5F5F5
White	#FFFFFF

Key Screens

Screen	Requirements
Splash Screen	App logo with 2-second delay

Screen	Requirements
Baby Registration	Form with baby details
Dashboard	Growth summary + reminders
Growth Tracker	Weight/height input + charts
Vaccination Screen	Vaccine schedule list
Feeding Guide	Nutrition information
Milestone Screen	Checklist interface
Settings/Profile	User preferences

10. Requirements Traceability Matrix (RTM)

FR ID	UC ID	Design Component	Test Case
FR-SSH-01	UC-SSH-01	BabyRegistrationScreen + BabyViewModel	TC-SSH-01
FR-SSH-05	UC-SSH-02	GrowthTrackerScreen + GrowthViewModel	TC-SSH-05
FR-SSH-07	UC-SSH-03	AnalyticsScreen + Chart Library	TC-SSH-07
FR-SSH-09	UC-SSH-04	VaccineScheduler	TC-SSH-09
FR-SSH-11	UC-SSH-05	NotificationManager	TC-SSH-11
FR-SSH-14	UC-SSH-07	MilestoneScreen	TC-SSH-14

11. Acceptance Criteria and Sign-off

11.1 Acceptance Checklist

Check	Pass
All FRs documented	Yes
All Use Cases completed	Yes
Database schema defined	Yes
RTM completed	Yes
NFRs measurable	Yes

11.2 Sign-off

Role	Name	Signature
Project Sponsor / Mentor	MindMatrix	
Developer	Mehek Munir Ahmed Tanga	Mehek Munir Ahmed Tanga
UI Developer	Mehek Munir Ahmed Tanga	Mehek Munir Ahmed Tanga
DB Handler	Mehek Munir Ahmed Tanga	

Post-Walkthrough Notes and Required Changes

[Record any changes required by the mentor before final acceptance. Format: 'UC-SS-03 Exception Flow Ex-4: [Mentor instruction]'. Date each note. Resolve all noted changes before beginning Room DB coding.]

Software Requirements Document Template — MindMatrix Industry Readiness Programme — Android Internship. PRD references refer to SpendSense / EPT PRDs (Annex A). Version 1.0.